Daily Quant Digest: 2026-10-14

Today's topics

1. Probability & Statistics: Conditional expectation as a projection, the tower property, and the law of total variance (random sums)
2. Derivatives: Treasury futures: conversion factors, cheapest-to-deliver, the basis and the delivery options
3. Algorithms & Data Structures: Monotonic stacks ("Largest Rectangle in Histogram")

Difficulty today: Medium / Medium / Medium-Hard. The probability section is where measure theory starts paying rent; the derivatives section builds directly on yesterday's Treasury quoting and accrued interest.

---

1. Probability & Statistics: Conditional Expectation, the Tower Property, Total Variance

1.1 What $\mathbb{E}[X \mid Y]$ actually is

Undergraduate courses define $\mathbb{E}[X \mid Y = y]$ as a number: the mean of X under the conditional distribution given $Y = y$. The grown-up object is $\mathbb{E}[X \mid Y]$, which is a **random variable**: take the function $g(y) = \mathbb{E}[X \mid Y = y]$ and plug in Y. So $\mathbb{E}[X \mid Y] = g(Y)$.

More generally, for a sigma-algebra $\mathcal{G}$ (think: "the information available"), $\mathbb{E}[X \mid \mathcal{G}]$ is the unique (up to null sets) random variable Z such that

1. Z is $\mathcal{G}$-measurable (you can compute it from the information in $\mathcal{G}$), and
2. $\mathbb{E}[Z\,\mathbf{1}_G] = \mathbb{E}[X\,\mathbf{1}_G]$ for every event $G \in \mathcal{G}$.

Condition 2 says: on every event you can see, Z has the same average as X. Existence comes from the Radon-Nikodym theorem; for interviews you never need that, but you do need the next viewpoint.

1.2 The projection picture

Work in L^2, the space of random variables with finite variance, with inner product $\langle U, V \rangle = \mathbb{E}[UV]$. The set of all $\mathcal{G}$-measurable square-integrable variables is a closed subspace. Then:

> **Theorem.** $\mathbb{E}[X \mid \mathcal{G}]$ is the orthogonal projection of X onto that subspace. Equivalently, it is the unique minimizer of $\mathbb{E}[(X - Z)^2]$ over all $\mathcal{G}$-measurable Z.

Proof sketch. Let $\hat{X} = \mathbb{E}[X \mid \mathcal{G}]$. The defining property extends from indicators $\mathbf{1}_G$ to any bounded $\mathcal{G}$-measurable W (by linearity and limits), so $\mathbb{E}[(X - \hat{X})W] = 0$: the error is orthogonal to everything you can see. Then for any $\mathcal{G}$-measurable Z,

$$\mathbb{E}[(X - Z)^2] = \mathbb{E}[(X - \hat{X})^2] + \mathbb{E}[(\hat{X} - Z)^2] + 2\,\mathbb{E}[(X - \hat{X})(\hat{X} - Z)].$$

The cross term is zero because $\hat{X} - Z$ is $\mathcal{G}$-measurable. So $\mathbb{E}[(X - Z)^2] \geq \mathbb{E}[(X - \hat{X})^2]$ with equality iff $Z = \hat{X}$. This is Pythagoras.

Why a quant should care:

- **Best predictor.** $\mathbb{E}[X \mid \mathcal{G}]$ is the best mean-squared-error forecast of X given information $\mathcal{G}$. Linear regression is the projection onto the smaller subspace of *linear* functions of the features, so OLS can only be worse (or equal). The gap between them is exactly the nonlinearity you are leaving on the table.
- **Pricing.** An arbitrage-free price process is $V_t = \mathbb{E}^{\mathbb{Q}}[\,\text{discounted payoff} \mid \mathcal{F}_t]$: today's price is the projection of the payoff onto today's information.

1.3 The rules you must be able to use without thinking

For integrable X, Y and sigma-algebras $\mathcal{H} \subseteq \mathcal{G}$:

Rule	Statement	Intuition
Linearity	$\mathbb{E}[aX + bY \mid \mathcal{G}] = a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}]$	projections are linear
Taking out what is known	if W is $\mathcal{G}$-measurable, $\mathbb{E}[WX \mid \mathcal{G}] = W\,\mathbb{E}[X \mid \mathcal{G}]$	W is a constant once you know $\mathcal{G}$
Independence	if $X \perp \mathcal{G}$, $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$	the information is useless
Tower	$\mathbb{E}[\,\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}]$	projecting onto a big subspace then a smaller one is the same as projecting onto the smaller one directly
Special case	$\mathbb{E}[\,\mathbb{E}[X \mid Y]\,] = \mathbb{E}[X]$	this is the law of total expectation

Tower proof sketch. For $H \in \mathcal{H} \subseteq \mathcal{G}$: $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]\mathbf{1}_H] = \mathbb{E}[X\mathbf{1}_H]$ because $H \in \mathcal{G}$. So $\mathbb{E}[X \mid \mathcal{G}]$ and X have the same averages on every $H \in \mathcal{H}$, which is the definition of having the same $\mathcal{H}$-conditional expectation.

Trading reading of the tower property: "your expected future forecast equals your current forecast." If you could predict that tomorrow you will revise your view up, you should revise it up today. That is exactly the martingale property of V_t above.

1.4 The law of total variance

$$\operatorname{Var}(X) = \underbrace{\mathbb{E}[\operatorname{Var}(X \mid Y)]}_{\text{unexplained (within)}} + \underbrace{\operatorname{Var}(\mathbb{E}[X \mid Y])}_{\text{explained (between)}}$$

Proof. Let $\hat{X} = \mathbb{E}[X \mid Y]$. Write $X - \mathbb{E}X = (X - \hat{X}) + (\hat{X} - \mathbb{E}X)$. The two pieces are orthogonal (the second is a function of Y), so

$$\operatorname{Var}(X) = \mathbb{E}[(X - \hat{X})^2] + \mathbb{E}[(\hat{X} - \mathbb{E}X)^2].$$

The second term is $\operatorname{Var}(\hat{X})$ since $\mathbb{E}\hat{X} = \mathbb{E}X$ by the tower rule. The first is $\mathbb{E}[\,\mathbb{E}[(X - \hat{X})^2 \mid Y]\,] = \mathbb{E}[\operatorname{Var}(X \mid Y)]$. Done: it is Pythagoras again.

Consequences:

- $\operatorname{Var}(\mathbb{E}[X \mid Y]) \le \operatorname{Var}(X)$: forecasts are less volatile than outcomes. A forecaster whose predictions are more volatile than the target is overfitting.
- The ratio $\operatorname{Var}(\mathbb{E}[X \mid Y])/\operatorname{Var}(X)$ is the population R^2 of the best (nonlinear) predictor, sometimes called the correlation ratio η^2.
- **Mixtures fatten tails.** If returns are $\mathcal{N}(0, \sigma^2)$ given a random vol σ, then $\operatorname{Var}(R) = \mathbb{E}[\sigma^2]$, but the kurtosis is $3\,\mathbb{E}[\sigma^4]/(\mathbb{E}[\sigma^2])^2 \ge 3$ by Jensen. Stochastic vol is the reason the FX smile has wings.

1.5 Random sums (Wald and the compound formula)

Let $N \ge 0$ be an integer random variable independent of i.i.d. $X_1, X_2, \ldots$ with mean μ and variance σ^2, and let $S = \sum_{i=1}^{N} X_i$. Conditioning on N:

- $\mathbb{E}[S \mid N] = N\mu$, so $\mathbb{E}[S] = \mu\,\mathbb{E}[N]$.
- $\operatorname{Var}(S \mid N) = N\sigma^2$, so

$$\operatorname{Var}(S) = \mathbb{E}[N]\,\sigma^2 + \operatorname{Var}(N)\,\mu^2.$$

The first term is noise *within* a given number of terms; the second is the uncertainty about *how many* terms. Insurance losses, daily P&L as (number of trades) times (P&L per trade), and the compound Poisson jump component in Merton's model all use this formula. For $N \sim \text{Poisson}(\lambda)$ it collapses to $\operatorname{Var}(S) = \lambda(\sigma^2 + \mu^2) = \lambda\,\mathbb{E}[X^2]$.

1.6 Pitfalls

1. **Treating $\mathbb{E}[X \mid Y]$ as a number.** It is a function of Y. Writing $\operatorname{Var}(\mathbb{E}[X \mid Y])$ only makes sense with the random-variable view.
2. **Forgetting the second variance term.** $\operatorname{Var}(S) = \mathbb{E}[N]\sigma^2$ is the most common wrong answer in random-sum questions; it ignores the randomness of N.

3. **Independence assumptions.** The random-sum formulas need N independent of the X_i (or at least N a stopping time for the mean, which is Wald's identity, coming later in the curriculum). If you stop rolling "when you get a 6", N depends on the X_i and the naive variance formula fails.
4. **Conditioning on a null event.** For continuous Y, $\mathbb{E}[X \mid Y = y]$ is only defined up to null sets, and naive limits can depend on how you approach the event (the Borel-Kolmogorov paradox). Always say which variable you condition on.
5. **Jensen in the wrong direction.** $\mathbb{E}[1/X \mid Y] \neq 1/\mathbb{E}[X \mid Y]$ in general; conditional Jensen gives $\geq$ for convex functions.

1.7 Interview question

> You roll a fair six-sided die and get N. You then roll N more fair dice and let S be their total.
>
> a. Find $\mathbb{E}[S]$ and $\mathrm{Var}(S)$.
>
> b. I tell you $S = 2$. What is $\mathbb{E}[N \mid S = 2]$?
>
> c. Follow-up (SIG / Optiver style): a stick of length 1 is broken at a uniform point and you keep the left piece; that piece is broken at a uniform point and you keep the left piece again. Find the mean and variance of what you are left with.

1.8 Solution

Building blocks. A single fair die X has

$$\mu = \mathbb{E}X = \tfrac{1+2+\cdots+6}{6} = \tfrac{21}{6} = 3.5, \qquad \mathbb{E}X^2 = \tfrac{1+4+9+16+25+36}{6} = \tfrac{91}{6},$$

$$\sigma^2 = \tfrac{91}{6} - \tfrac{49}{4} = \tfrac{182-147}{12} = \tfrac{35}{12} \approx 2.9167.$$

N is itself a die, so $\mathbb{E}N = 3.5$ and $\mathrm{Var}(N) = 35/12$.

(a) Mean. Conditioning on N: $\mathbb{E}[S \mid N] = 3.5N$. Tower property:

$$\mathbb{E}[S] = \mathbb{E}[3.5N] = 3.5 \times 3.5 = 12.25.$$

(a) Variance. Given N, S is a sum of N independent dice, so $\mathrm{Var}(S \mid N) = N \cdot \frac{35}{12}$. Law of total variance:

$$\mathrm{Var}(S) = \mathbb{E}\left[\tfrac{35}{12}N\right] + \mathrm{Var}(3.5N) = \tfrac{35}{12}(3.5) + (3.5)^2 \cdot \tfrac{35}{12}.$$

Factor out $\frac{35}{12}$:

$$\mathrm{Var}(S) = \tfrac{35}{12}\left(3.5 + 12.25\right) = \tfrac{35}{12} \times 15.75 = \tfrac{735}{16} = 45.9375.$$

The decomposition is instructive: the within term is 10.21 and the between term is 35.73. About 78% of the variance comes from not knowing *how many* dice you roll, not from the dice themselves. Standard deviation ≈ 6.78.

(b) Bayes on top of conditioning. $S = 2$ is only possible with $N = 1$ (roll a 2) or $N = 2$ (roll 1 and 1).

$$\mathbb{P}(N = 1, S = 2) = \tfrac{1}{6} \cdot \tfrac{1}{6} = \tfrac{1}{36} = \tfrac{6}{216}, \qquad \mathbb{P}(N = 2, S = 2) = \tfrac{1}{6} \cdot \tfrac{1}{36} = \tfrac{1}{216}.$$

So the posterior odds are $6 : 1$ and

$$\mathbb{E}[N \mid S = 2] = \frac{1 \cdot 6 + 2 \cdot 1}{6 + 1} = \frac{8}{7} \approx 1.143.$$

Note how different this is from the prior mean of 3.5: a small total is strong evidence of few dice.

(c) The broken stick. Let $L_1 \sim U(0, 1)$ and $L_2 \mid L_1 \sim U(0, L_1)$.

- $\mathbb{E}[L_2 \mid L_1] = L_1/2$, so $\mathbb{E}L_2 = \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{4}$.
- $\mathrm{Var}(L_2 \mid L_1) = L_1^2/12$ (variance of a uniform on an interval of length L_1), so the within term is $\mathbb{E}[L_1^2]/12 = \tfrac{1}{3} \cdot \tfrac{1}{12} = \tfrac{1}{36}$.
- The between term is $\mathrm{Var}(L_1/2) = \tfrac{1}{4} \cdot \tfrac{1}{12} = \tfrac{1}{48}$.
- Total: $\mathrm{Var}(L_2) = \tfrac{1}{36} + \tfrac{1}{48} = \tfrac{4+3}{144} = \tfrac{7}{144} \approx 0.0486$.

Interview trick for (c). Write $L_2 = U_1 U_2$ with U_1, U_2 i.i.d. uniform (break at fraction U_2 of the current piece). Independence gives $\mathbb{E}L_2 = \tfrac{1}{4}$ and $\mathbb{E}L_2^2 = \mathbb{E}U_1^2 \, \mathbb{E}U_2^2 = \tfrac{1}{9}$, so $\mathrm{Var} = \tfrac{1}{9} - \tfrac{1}{16} = \tfrac{7}{144}$. Same answer in two lines, and it generalizes: after k breaks the mean is 2^{-k} and the second moment is 3^{-k}.

Sanity checks.

- S ranges from 1 to 36 with mean 12.25; a standard deviation of about 6.8 is plausible for such a wide, right-skewed range.
- If N were fixed at 3.5 (impossible, but as a thought experiment), variance would be only $3.5 \times 35/12 \approx 10.2$. The random N must add variance, and it does.
- A one-million-path simulation gives mean 12.26 and variance 45.95, matching 12.25 and 45.9375.

---

2. Derivatives: Treasury Futures, Conversion Factors and Cheapest-to-Deliver

2.1 The contract

The CME 10-Year T-Note future (ticker ZN, often called "TY" on Bloomberg) is the most traded rates futures contract in the world. Key terms:

- **Notional:** $100,000 face value of Treasury notes.
- **Deliverable grade:** any US Treasury note with remaining maturity of at least 6.5 years and less than 8 years, measured from the first day of the delivery month. The seller (short) chooses which one to deliver.
- **Quote:** points and 32nds of par, with half-32nd ticks: "111-14+" means $111 + 14.5/32 = 111.453125$. One full 32nd is $31.25 per contract; the minimum tick of half a 32nd is $15.625.
- **Delivery months:** March, June, September, December. Delivery can happen on any business day of the delivery month, at the short's choice. Trading stops seven business days before the last business day of the month (for December 2026: around December 21), but delivery can still occur until the last business day (December 31, 2026).
- **Settlement:** physical. The short delivers bonds through Fedwire and receives the **invoice price**.

The other tenors (2Y ZT, 5Y ZF, Ultra-10 TN, Bond ZB, Ultra Bond UB) work the same way with different baskets.

2.2 Why a basket, and why conversion factors

A single-bond contract would be easy to squeeze, so the exchange lets many bonds be delivered. But a 4.5% note is worth more than a 3.5% note, so the short must be paid more for delivering it. The **conversion factor** (CF) does this scaling:

$$\text{Invoice price} = F \times \text{CF} + \text{AI}_{\text{delivery}}$$

per 100 face, where F is the futures settlement price and AI is accrued interest at delivery.

Definition. CF is the clean price (per $1 face) at which the bond would yield exactly **6%** (semiannual), with its maturity truncated to whole quarters (for ZN, ZB) measured from the first day of the delivery month. The CME formula, with coupon c in decimal:

- n = whole years to maturity, z = remaining months rounded down to a quarter (0, 3, 6 or 9)
- $v = z$ if $z < 7$, else $v = 3$
- $a = 1.03^{-v/6},\ b = \frac{c}{2} \cdot \frac{6-v}{6},\ C = 1.03^{-2n}$ if $z < 7$ else $1.03^{-(2n+1)},\ d = \frac{c}{0.06}(1 - C)$
- $\text{CF} = a\left(\frac{c}{2} + C + d\right) - b$, rounded to 4 decimals.

The intuition: a coupon below 6% means the bond would trade below par at a 6% yield, so CF < 1. Today, with yields around 4%, every deliverable has CF around 0.87 to 0.91, and the futures price sits around $100/0.9 \approx 111$.

2.3 Cheapest-to-deliver

At delivery, the short buys a bond in the cash market for $P + \text{AI}$ and receives $F \cdot \text{CF} + \text{AI}$. Accrued cancels, so the short's profit from delivering bond i is $F \cdot \text{CF}_i - P_i$. The short picks the bond that maximizes this, i.e. minimizes

$$\text{gross basis}_i = P_i - F \cdot \text{CF}_i \qquad \Longleftrightarrow \qquad \text{minimizes } \frac{P_i}{\text{CF}_i}.$$

At expiry no-arbitrage forces $F = \min_i P_i/\text{CF}_i$, so the CTD has zero basis at delivery. Before delivery, three numbers matter:

- **Gross basis** $= P - F \cdot \text{CF}$, quoted in 32nds.
- **Carry** = coupon income − repo financing cost over the holding period.
- **Net basis** = gross basis − carry. This is the price of the short's delivery options.
- **Implied repo rate (IRR)**: the return from buying the bond, repo-ing it, selling futures and delivering:

$$\text{IRR} = \frac{(F \cdot \text{CF} + \text{AI}_{\text{del}} + \text{coupons}) - (P + \text{AI}_{\text{now}})}{P + \text{AI}_{\text{now}}} \cdot \frac{360}{\text{days}}.$$

The pre-expiry CTD is the bond with the **highest IRR** (equivalently, lowest net basis). If IRR exceeds the actual repo rate you could earn riskless money, so in practice the best IRR sits slightly below repo.

Duration rule of thumb. Conversion factors price every bond at a 6% yield. When market yields are below 6%, long-duration bonds have rallied more above their 6% value than short-duration bonds, so their P/CF is larger and they are expensive to deliver: the **shortest-duration** bond (short maturity, high coupon) tends to be CTD. When yields are above 6%, the ordering flips and the **longest-duration** bond tends to be CTD. As yields move, the CTD can switch, which is the source of the futures' negative convexity.

2.4 Worked scenario: finding the CTD for TYZ6

Trade date: Wednesday 14 Oct 2026, cash settlement Thursday 15 Oct. Delivery assumed on the last business day, Thursday 31 Dec 2026 (77 days). Term repo: 3.75% (act/360). Futures: TYZ6 at **111-14+** $= 111.453125$. The basket below uses three illustrative notes (chosen so that no coupon falls between settlement and delivery, which keeps the arithmetic clean):

Note	Clean price	Yield	Time from 1 Dec 2026	n, z	CF
A: 4.125% Aug-31-2033	100-14 (100.4375)	4.051%	6y 8m 30d	6, 6	0.9003
B: 4.000% Feb-15-2034	99-02 (99.0625)	4.149%	7y 2m 14d	7, 0	0.8870
C: 3.875% Aug-15-2034	97-27 (97.84375)	4.200%	7y 8m 14d	7, 6	0.8732

Step 1: conversion factor for A ($c = 0.04125$, $n = 6$, $z = 6$, so $v = 6$):

- $a = 1.03^{-1} = 0.970874$, $b = 0.020625 \times 0/6 = 0$
- $C = 1.03^{-12} = 0.701380$
- $d = (0.04125/0.06)(1 - 0.701380) = 0.6875 \times 0.298620 = 0.205301$
- $\text{CF} = 0.970874 \times (0.020625 + 0.701380 + 0.205301) - 0 = 0.970874 \times 0.927306 = 0.9003$.

Sanity check: CF should equal the 6%-yield clean price of a 4.125% bond with exactly 6.5 years left, which is 0.9003 (it is).

Step 2: accrued interest (actual/actual, from yesterday's digest). A's last coupon was 31 Aug 2026 and the next is 28 Feb 2027 (181-day period). B and C last paid on 15 Aug 2026, next 15 Feb 2027 (184 days).

	AI on 15 Oct (per 100)	AI on 31 Dec (per 100)	Coupon income ΔAI
A	$2.0625 \times 45/181 = 0.5128$	$2.0625 \times 122/181 = 1.3902$	0.8774
B	$2.0000 \times 61/184 = 0.6630$	$2.0000 \times 138/184 = 1.5000$	0.8370
C	$1.9375 \times 61/184 = 0.6423$	$1.9375 \times 138/184 = 1.4531$	0.8108

Step 3: gross basis, carry, net basis, IRR. Financing cost $= (P + \text{AI}_{\text{now}}) \times 0.0375 \times 77/360$.

	$F \cdot \text{CF}$	Gross basis	Financing	Carry	Net basis	IRR
A	100.3412	0.0963 (3.1/32)	0.8097	+0.0677 (2.2/32)	**0.0285 (0.9/32)**	**3.62%**
B	98.8589	0.2036 (6.5/32)	0.7999	+0.0371 (1.2/32)	0.1665 (5.3/32)	2.97%
C	97.3209	0.5229 (16.7/32)	0.7899	+0.0209 (0.7/32)	0.5020 (16.1/32)	1.37%

Note A is the CTD: lowest net basis, highest implied repo (3.62%, just below the 3.75% repo rate, as it should be). It is also the shortest-duration note, exactly what the "yields below 6%" rule predicts.

Step 4: invoice amount if A is delivered at 111-14+ on 31 Dec:

$$\left(111.453125 \times 0.9003 + 1.3902\right) \times 1000 = 100{,}341.25 + 1{,}390.20 = \$101{,}731.44 \text{ per contra}$$

Step 5: a CTD switch. Shift all three yields in parallel and recompute P/CF (the at-delivery CTD criterion):

P/CF	**Yields** -100**bp**	**Today**	**Yields** $+100$**bp**
A (short)	**118.42**	**111.56**	105.16
B	118.98	111.68	104.90
C (long)	119.84	112.05	**104.85**

A 100bp selloff hands the CTD to the longest note C. The futures then start tracking a longer-duration bond, so the contract's DV01 rises in a selloff and falls in a rally less than a bond's would: negative convexity, which is the short's option.

Step 6: hedging with futures. The futures behaves like the CTD scaled by $1/\mathrm{CF}$:

$$\mathrm{DV01}_{\mathrm{fut}} \approx \frac{\mathrm{DV01}_{\mathrm{CTD}}}{\mathrm{CF}_{\mathrm{CTD}}} = \frac{\$59.65}{0.9003} \approx \$66.26 \text{ per contract.}$$

To hedge \$50M face of note B (DV01 \$62.35 per \$100k, so \$31,175 total), sell $31{,}175/66.26 \approx 470.5$, so about **470 to 471 contracts**.

Step 7: the basis trade. "Buying the basis" means buying the cash bond and selling $\mathrm{CF}\times$ face/100k futures. Buying \$10M of A and selling $100 \times 0.9003 \approx 90$ contracts, you pay the gross basis of 3.1/32 (about \$9,630) and earn carry of 2.2/32 (about \$6,770). If nothing moves, your expected loss is the net basis, 0.9/32 or about **\$2,850**. That is the premium you pay for being long the short's delivery options: if yields jump 100bp and C becomes CTD, the futures underperform A and your short futures leg makes money.

2.5 The delivery options (what the net basis buys)

1. **Quality (switch) option:** the right to deliver whichever bond is cheapest at the time. Worth most when two bonds are close to each other in P/CF and yields are volatile.
2. **Timing option:** delivery on any day of the month. With positive carry (coupon > repo), the short holds to the last day; with negative carry, delivers early.
3. **Wild card option:** for ZB and similar contracts, futures settle at 2pm Chicago time but the short can declare intention to deliver until 6pm at that day's settlement price. A post-close move in cash bonds can be locked in.

4. **End-of-month option:** after trading stops, the invoice price is fixed but the short still has a week to choose which bond and when.

2.6 What can go wrong

- **CTD switch risk:** a hedge sized off the current CTD's DV01 becomes wrong after a large move. Desks hedge with an option-adjusted futures DV01.
- **Repo squeezes:** if the CTD goes "special" in repo, its carry changes and the basis can blow out. The March 2020 basis trade unwind (hedge funds long cash, short futures, financed in repo) is the canonical example of the basis trade turning into a liquidity crisis.
- **Delivery logistics:** if you are short at the end of the month without the bond, you must buy it. Being caught short in a squeeze can cost multiple points.
- **Rounding:** CF uses truncated maturity, so two very similar bonds can get noticeably different factors. This shows up in which bond is CTD.

2.7 Likely interview questions

1. **"Why is the shortest-duration bond usually CTD when yields are below 6%?"** CF prices every bond at 6%. Below 6%, long-duration bonds have risen more in price relative to their 6% value than short ones, so P/CF is larger for long duration. The short delivers the bond with the smallest P/CF, which is the low-duration one. Above 6% the ordering flips.
2. **"What is the implied repo rate and why is it bounded by actual repo?"** It is the financing rate implied by buying the cash bond and delivering it into a short futures position. If IRR > repo you would buy bonds, finance in repo, sell futures and lock in profit; that pushes IRR down. The gap between them, in price terms, is the net basis, i.e. the value of the delivery options.
3. **"How many 10Y futures to hedge $100M of a 10-year bond with DV01 of $85 per $100k?"** Bond DV01 is $85,000. Futures DV01 $\approx$ CTD DV01 / CF; with the numbers above, $66.26. So about $85{,}000/66.26 \approx 1{,}283$ contracts short.

---

3. Algorithms & Data Structures: Monotonic Stacks ("Largest Rectangle in Histogram")

3.1 Problem

Given an array `heights` of non-negative integers, where each bar has width 1, return the area of the largest rectangle that fits entirely inside the histogram.

Constraints: $1 \le n \le 10^5$, $0 \le \text{heights}[i] \le 10^4$.

Examples:

- `[2,1,5,6,2,3]` returns `10` (the bars 5 and 6, height 5, width 2).
- `[2,4]` returns `4`.
- `[6,2,5,4,5,1,6]` returns `12` (bars 5,4,5 at height 4, width 3).

Market flavour: replace bars by a liquidity profile per price level; the problem is "the largest block of size you can trade across consecutive levels at uniform depth".

3.2 Brute force

Every optimal rectangle has a height equal to the shortest bar it spans (otherwise you could make it taller). So try every pair (l, r) and take $\min(h[l..r]) \cdot (r - l + 1)$. Maintaining the running minimum as you extend r gives $O(n^2)$ time and $O(1)$ space. At $n = 10^5$ that is 5×10^9 steps, too slow.

3.3 The insight: each bar as the limiting height

Flip the question. For each bar i, ask: what is the widest rectangle whose height is exactly $h[i]$? It extends left until the first bar shorter than $h[i]$ (index L_i) and right until the first bar shorter than $h[i]$ (index R_i). Its area is $h[i] \cdot (R_i - L_i - 1)$. The answer is the max over i. So the problem reduces to: **for every element, find the nearest smaller element on each side.** That is the textbook job of a monotonic stack.

Monotonic stack. Sweep left to right, keeping a stack of indices whose heights are strictly increasing from bottom to top. When bar i arrives with height h:

- While the top of the stack has height $\geq h$, pop it. For the popped bar, i is the first bar on its right that is not taller, so $R = i$. And the element now *under* it on the stack is the first bar on its left that is smaller, so L = new top (or -1 if the stack is empty). Compute its area.
- Push i.

At the end, push a virtual bar of height 0 at index n to flush the stack.

Why it is correct. The stack is always increasing, so whatever sits beneath an index is the nearest smaller-height bar to its left (everything in between was popped by something no taller than it). When an index is popped by bar i, nothing between it and i was shorter (those would have popped it earlier).

Ties. Popping on $\geq$ means that with equal heights, the earlier bar is popped with a width that stops short of its twin. That is fine: the *last* bar of an equal run gets the full width, so the maximum is still found. The random cross-check in the code confirms this.

Complexity. Each index is pushed once and popped once, so $O(n)$ time total despite the nested loop (amortized analysis). Space $O(n)$ for the stack.

Trace on `[2,1,5,6,2,3]`:

i	h	Pops (index: area)	Stack after
0	2		[0]
1	1	0: $2 \times (1 - (-1) - 1) = 2$	[1]
2	5		[1,2]
3	6		[1,2,3]
4	2	3: $6 \times (4 - 2 - 1) = 6$; 2: $5 \times (4 - 1 - 1) = 10$	[1,4]
5	3		[1,4,5]
6	0	5: $3 \times 1 = 3$; 4: $2 \times 4 = 8$; 1: $1 \times 6 = 6$	[6]

Maximum is 10.

3.4 C++ solution

The full file `2026-10-14.cpp` includes this function plus tests: the examples, a 10^5-bar stress case, and 2000 random arrays checked against the brute force.

```cpp
#include <algorithm>
#include <vector>

long long largestRectangleArea(const std::vector<int>& heights) {
    const int n = static_cast<int>(heights.size());
    std::vector<int> stk;          // indices with strictly increasing heights
    stk.reserve(n + 1);
    long long best = 0;

    for (int i = 0; i <= n; ++i) {
        const int h = (i == n) ? 0 : heights[i];   // sentinel at i == n
        while (!stk.empty() && heights[stk.back()] >= h) {
            const int top = stk.back();
            stk.pop_back();
            const int left = stk.empty() ? -1 : stk.back();
            const long long width = i - left - 1;
            best = std::max(best, static_cast<long long>(heights[top]) * width);
        }
        stk.push_back(i);
    }
    return best;
}
```

Output of the test harness:

```
[2,1,5,6,2,3] -> 10  OK
[2,4] -> 4  OK
[1] -> 1  OK
[0,0,0] -> 0  OK
[3,3,3,3] -> 12  OK
[1,2,3,4,5] -> 9  OK
[5,4,3,2,1] -> 9  OK
[6,2,5,4,5,1,6] -> 12  OK
```

```
[10000,10000,10000] -> 30000  OK
100000 x 10000 -> 1000000000
2000 random tests vs brute force: all passed
```

3.5 Walkthrough of the C++

- `const std::vector<int>& heights`: pass by const reference. A by-value parameter would copy 10^5 ints on every call; `&` avoids the copy and `const` promises not to modify the caller's data.
- `std::vector<int>` **as a stack**: `std::stack` exists, but it is just an adapter over `std::deque` by default and hides `reserve`. A `vector` with `push_back`, `back` and `pop_back` is the idiomatic, cache-friendly stack in competitive and low-latency code.
- `stk.reserve(n + 1)`: allocates capacity once, so `push_back` never reallocates. Without it the vector would grow geometrically (amortized $O(1)$, but with a few reallocations and copies).
- **Storing indices, not heights**: you need the index to compute width and can always recover the height with `heights[idx]`. This is the standard monotonic-stack pattern.
- **The sentinel** `i == n`: the loop runs one extra step with `h = 0`, which pops everything left. This removes a separate cleanup loop after the main loop. Note we never read `heights[n]`, which would be undefined behaviour; the ternary guards it.
- `!stk.empty() && heights[stk.back()] >= h`: `&&` short-circuits, so `stk.back()` is never called on an empty vector (that would also be undefined behaviour).
- `long long` **and** `static_cast`: the maximum area is $10^4 \times 10^5 = 10^9$, which fits in a 32-bit `int` (max about 2.1×10^9), but only just. Casting one operand to `long long` before multiplying makes the product 64-bit; casting *after* the multiplication would be too late, because the overflow would already have happened in `int`.
- `std::max(best, ...)`: both arguments must have the same type (it is a template deduced from both arguments), which is another reason for the explicit cast.
- **In the test file**: `for (const auto& [h, expected] : cases)` is a C++17 *structured binding*: it unpacks each `Case` struct into named references without copying. `std::mt19937` with a fixed seed plus `std::uniform_int_distribution` gives reproducible random tests, and `assert` aborts loudly on a mismatch (it is compiled out if you define `NDEBUG`).

3.6 Variations you should be ready for

- **Maximal rectangle of 1s in a binary matrix:** build a histogram per row (height = consecutive 1s above) and run this algorithm on each row: $O(rows \times cols)$.
- **Next greater element / daily temperatures / stock span:** the same stack with a different pop condition.
- **Sum of subarray minimums:** $h[i]$ is the minimum of exactly $(i - L_i)(R_i - i)$ subarrays, so the answer is $\sum_i h[i](i - L_i)(R_i - i)$; use strict inequality on one side and non-strict on the other to avoid double counting ties.
- **Trapping rain water:** solvable with a monotonic (decreasing) stack or with two pointers.